SIMATS ENGINEERING

ASSESSMENT TOOL 2 - SCENARIO BASED QUESTIONS

Course Name	Computer Networks
Course Code	CSA0709
Course Outcome (CO2)	Configure IP addressing schemes, subnetting, and basic routing techniques for efficient network design and topology (BL3)
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Assessment Tool Weightage	30%

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Question 1 (10 Marks)

Scenario: A Smart Campus is allocated 192.168.100.0/24. Subnet masks assigned: School of Computing = /25, School of Electronics = /26, School of Mechanical Engineering = /27. Determine (1) network and broadcast address, (2) valid host range, (3) usable hosts of each subnet, and (4) the remaining unused range.

1. Understanding of the Scenario

- Parent block 192.168.100.0/24 has 256 addresses; three departments have fixed masks /25, /26 and /27.
- Required: network and broadcast addresses, host ranges, usable hosts, and the leftover address range.

2. Identification of Key Issues

- Subnets must not overlap and each must start on a multiple of its block size.
- Network and broadcast addresses cannot be assigned to hosts, so usable hosts = block size - 2.
- Largest subnet is allocated first so the free space stays in one continuous block.

3. Application of Concepts (Solution)

Formulae: block size = $2^{(32 - \text{prefix})}$; usable hosts = block size - 2; network = first address; broadcast = last address.

Department	Prefix	Mask	Block size	Usable hosts
Computing	/25	255.255.255.128	128	128 - 2 = 126
Electronics	/26	255.255.255.192	64	64 - 2 = 62
Mechanical	/27	255.255.255.224	32	32 - 2 = 30

Allocation: Computing .0 to .127; Electronics starts at next free address .128 and ends at .191; Mechanical starts at .192 and ends at .223.

4. Final Answer

Department	Network Address	Broadcast	Valid Host Range	Usable Hosts
Computing (/25)	192.168.100.0	192.168.100.127	192.168.100.1 - 192.168.100.126	126
Electronics (/26)	192.168.100.128	192.168.100.191	192.168.100.129 - 192.168.100.190	62
Mechanical (/27)	192.168.100.192	192.168.100.223	192.168.100.193 - 192.168.100.222	30

Unused range: 192.168.100.224 - 192.168.100.255 (32 addresses = 256 - 224 used; this equals one free /27 block, 192.168.100.224/27).

5. Decision Making and Conclusion

The allocation is valid and overlap-free: 128 + 64 + 32 = 224 addresses are used, so the Smart Campus keeps one /27 (30 usable hosts) in reserve for a new department. Total usable hosts across the three departments = 126 + 62 + 30 = 218.

Question 2 (10 Marks)

Scenario: A software company is assigned the private block 10.10.0.0/16. Masks: Software Development = /24, Quality Assurance = /25, Human Resources = /26, Executive Management = /27. Determine (1) network and broadcast address, (2) valid host range, (3) usable hosts, and (4) the unused range.

1. Understanding of the Scenario

- Parent block 10.10.0.0/16 has 65,536 addresses; four departments need subnets of /24, /25, /26 and /27.
- Required: network/broadcast addresses, host ranges, usable hosts and the unused range.

2. Identification of Key Issues

- Subnets must be aligned to their block sizes and must not overlap.
- The /24 consumes the whole 10.10.0.x range, so the next subnets start in 10.10.1.x.
- Each subnet loses two addresses (network and broadcast) from its host count.

3. Application of Concepts (Solution)

Formulae: block size = $2^{(32 - \text{prefix})}$; usable hosts = block size - 2.

Department	Prefix	Mask	Block size	Usable hosts
Software Dev.	/24	255.255.255.0	256	256 - 2 = 254
Quality Assurance	/25	255.255.255.128	128	128 - 2 = 126
Human Resources	/26	255.255.255.192	64	64 - 2 = 62
Executive Mgmt.	/27	255.255.255.224	32	32 - 2 = 30

Allocation: Software Dev. 10.10.0.0 - 10.10.0.255; QA 10.10.1.0 - 10.10.1.127; HR 10.10.1.128 - 10.10.1.191; Executive 10.10.1.192 - 10.10.1.223.

4. Final Answer

Department	Network Address	Broadcast	Valid Host Range	Usable Hosts
Software Dev. (/24)	10.10.0.0	10.10.0.255	10.10.0.1 - 10.10.0.254	254
QA (/25)	10.10.1.0	10.10.1.127	10.10.1.1 - 10.10.1.126	126
HR (/26)	10.10.1.128	10.10.1.191	10.10.1.129 - 10.10.1.190	62
Executive (/27)	10.10.1.192	10.10.1.223	10.10.1.193 - 10.10.1.222	30

Unused range: 10.10.1.224 - 10.10.255.255. Addresses used = 256 + 128 + 64 + 32 = 480; unused = 65,536 - 480 = **65,056**.

5. Decision Making and Conclusion

Every department gets a correctly sized, non-overlapping subnet, using only the first 480 addresses of the /16. Total usable hosts = 254 + 126 + 62 + 30 = 472. The large remaining range 10.10.1.224 - 10.10.255.255 gives plenty of room for expansion of the headquarters.

Question 3 (10 Marks)

Scenario: A smart hospital migrates from IPv4 to IPv6 and is assigned 2001:0db8:abcd:0000:0000:0000:0000:0000/64. (1) Convert the address to compressed notation. (2) Expand the compressed address to full form. (3) Identify the network prefix using /64. (4) Calculate the number of host addresses in the subnet. (5) Give the total number of possible host addresses in the network.

1. Understanding of the Scenario

- An IPv6 address has 128 bits (8 groups of 16 bits). The given prefix length /64 means the first 64 bits identify the network.
- The hospital needs a subnet large enough for thousands of medical devices and IoT sensors.

2. Identification of Key Issues

- Long runs of zeros make the notation hard to read and error-prone, so compression rules are needed.
- Correctly separating the 64-bit network part from the 64-bit host part decides the address capacity.
- IPv6 has no broadcast address, so no addresses are subtracted from the host count.

3. Application of Concepts (Solution)

Step 1 - Compression: Given: 2001:0db8:abcd:0000:0000:0000:0000:0000. Remove leading zeros: 2001:db8:abcd:0:0:0:0:0. Replace the run of five zero groups with "::" giving **2001:db8:abcd::** (written with prefix: 2001:db8:abcd::/64).

Step 2 - Expansion: 2001:db8:abcd:: has 3 visible groups, so "::" stands for 8 - 3 = 5 zero groups. Insert 0:0:0:0:0 and pad each group to 4 digits: **2001:0db8:abcd:0000:0000:0000:0000:0000** (same as the original).

Step 3 - Network prefix (/64): first 64 bits = first four groups = 2001:0db8:abcd:0000, i.e. **2001:db8:abcd:0::/64** (shortest form 2001:db8:abcd::/64). The remaining 64 bits (0000:0000:0000:0000) form the interface ID.

Step 4 and 5 - Host addresses: host bits = 128 - 64 = 64, so total addresses = 2^{64} = 18,446,744,073,709,551,616. (Points 4 and 5 of the question ask for the same value.)

4. Final Answer

Item	Result
Compressed address	2001:db8:abcd::
Expanded address	2001:0db8:abcd:0000:0000:0000:0000:0000
Network prefix (/64)	2001:db8:abcd::/64
Host bits	64
Total host addresses (4 and 5)	2^{64} = 18,446,744,073,709,551,616

5. Decision Making and Conclusion

A /64 subnet is the standard IPv6 LAN size and easily covers thousands of medical devices and IoT sensors - about 18.4 quintillion addresses - so the hospital never has to worry about address shortage or NAT inside this subnet. Compression is only a shorthand; the compressed and expanded forms represent exactly the same address.

Question 4 (10 Marks)

Scenario: An ISP core router has routing table entries 192.168.1.0/24, 192.168.2.0/24 and 192.168.3.0/24. A packet arrives with destination IP 192.168.2.45. Determine (1) the destination subnet, (2) the matching routing table entry, (3) the next-hop network/interface, and (4) the default route used when no entry matches.

1. Understanding of the Scenario

- The router forwards packets by comparing the destination IP address with its routing table entries.
- Three /24 customer networks are listed; the packet is destined to 192.168.2.45.

2. Identification of Key Issues

- Find which entry's network prefix matches the destination (longest-prefix match).
- Decide the outgoing interface/next hop for the matching entry.
- Decide what happens when no entry matches (default route).

3. Application of Concepts (Solution)

Step 1 - Subnet of the destination: mask /24 = 255.255.255.0. 192.168.2.45 AND 255.255.255.0 = 192.168.2.0. Network = 192.168.2.0/24, range 192.168.2.0 - 192.168.2.255 (host range .1 - .254, broadcast .255).

Step 2 - Compare with the table: the first 24 bits of the destination are 192.168.2.

Routing entry	Network range	Match with 192.168.2.45?
192.168.1.0/24	192.168.1.0 - 192.168.1.255	No (third octet 1 differs)
192.168.2.0/24	192.168.2.0 - 192.168.2.255	**Yes** (first 24 bits match)
192.168.3.0/24	192.168.3.0 - 192.168.3.255	No (third octet 3 differs)

Step 3 - Next hop: the packet is forwarded out of the interface / next-hop router associated with the 192.168.2.0/24 entry (the customer-2 network). The question gives no interface names, so the interface is shown by its entry; if the network is directly connected, the router delivers the packet straight to host 192.168.2.45 on that interface.

Step 4 - Default route: if no entry matched, the router would use the default route **0.0.0.0/0** (gateway of last resort) pointing to the upstream ISP/Internet next hop. It matches every address with the shortest prefix (/0), so it is chosen only when nothing more specific matches.

4. Final Answer

- Destination subnet: **192.168.2.0/24**.
- Matching routing table entry: **192.168.2.0/24** (only one entry matches).
- Next hop/interface: the **interface or next hop configured for 192.168.2.0/24**.
- Default route: **0.0.0.0/0** via the upstream ISP gateway (used only if no entry matches).

5. Decision Making and Conclusion

The packet is forwarded correctly because exactly one routing entry matches, so there is no ambiguity in the choice. The default route is a safety net: without it, packets for unknown destinations would be dropped, and with it they are passed on to the upstream provider.

Question 5 (10 Marks)

Scenario: Two departments are connected by a Layer-3 router: Administration LAN = 192.168.10.0/26 (gateway 192.168.10.1) and Finance LAN = 192.168.10.64/26 (gateway 192.168.10.65). Determine (1) network and broadcast address of each LAN, (2) valid host range, (3) suitable IP addresses for three computers in each LAN, and (4) the default gateway for each LAN.

1. Understanding of the Scenario

- Two separate /26 subnets (mask 255.255.255.192, 64 addresses each) are joined by a Layer-3 router.
- Each router interface acts as the gateway of its LAN; hosts need addresses in their own subnet.

2. Identification of Key Issues

- Hosts in different LANs can talk only through the router, so each host needs the correct default gateway.
- Host addresses must be unique, inside the valid range, and must not be the network, broadcast or gateway address.
- Usable hosts = 2^6 - 2 = 62 per LAN.

3. Application of Concepts (Solution)

Administration LAN (192.168.10.0/26): block size 64, so addresses .0 to .63. Network = 192.168.10.0; broadcast = 192.168.10.63; hosts .1 to .62.

Finance LAN (192.168.10.64/26): addresses .64 to .127. Network = 192.168.10.64; broadcast = 192.168.10.127; hosts .65 to .126.

Host addressing: .1 (Administration) and .65 (Finance) are already used by the router interfaces, so computers start from the next address in each LAN.

4. Final Answer

Parameter	Administration LAN	Finance LAN
Subnet / mask	192.168.10.0/26 (255.255.255.192)	192.168.10.64/26 (255.255.255.192)
Network address	192.168.10.0	192.168.10.64
Broadcast address	192.168.10.63	192.168.10.127
Valid host range	192.168.10.1 - 192.168.10.62	192.168.10.65 - 192.168.10.126
Computer 1	192.168.10.2	192.168.10.66
Computer 2	192.168.10.3	192.168.10.67
Computer 3	192.168.10.4	192.168.10.68
Default gateway	192.168.10.1	192.168.10.65

5. Decision Making and Conclusion

Each computer uses the router interface of its own LAN as default gateway (192.168.10.1 for Administration, 192.168.10.65 for Finance), with subnet mask 255.255.255.192. A packet from 192.168.10.2 to 192.168.10.66 is sent to the Administration gateway, routed by the Layer-3 router and delivered to the Finance LAN. Each LAN still has 62 usable addresses (59 free after the gateway and three computers), leaving room for growth.